whisper.cpp vs openai/whisper -- Measured Benchmark

base.en model, both engines, same machine (Apple Silicon Mac), same audio. Real measurements, not estimates.

Test audio: JFK inaugural clip, 11.0s, 22-word ground-truth transcript ("...ask not what your country can do for you, ask what you can do for your country.").
Noise conditions: clean, and three synthetically-degraded variants (additive white noise at 5dB, 0dB, -5dB SNR) to get differentiated accuracy data -- the clean clip alone produced 0% WER on both engines, which doesn't distinguish them.
Raw engine numbers: isolated single-shot runs via `/usr/bin/time -l` (exact OS-level CPU/memory accounting), not estimated.
Live relay numbers: measured through the actual WebSocket relay servers used by the app (`relays/whispercpp_relay.py`, `relays/openai_whisper_relay.py`), 3 repeated trials each, real-time audio pacing.

1. Accuracy (Word Error Rate against ground truth)

Condition	whisper.cpp	openai/whisper	Winner
Clean audio	**100.00% accuracy (0.00% WER)**	**100.00% accuracy (0.00% WER)**	Tie
5dB SNR (light noise)	**100.00% accuracy (0.00% WER)**	**100.00% accuracy (0.00% WER)**	Tie
0dB SNR (moderate noise)	**95.45% accuracy (4.55% WER, 1/22 words wrong)**	90.91% accuracy (9.09% WER, 2/22 words wrong)	**whisper.cpp**
-5dB SNR (severe noise, noise louder than speech)	63.64% accuracy (36.36% WER, 8/22 words wrong)	**77.27% accuracy (22.73% WER, 5/22 words wrong)**	**openai/whisper**

Not a clean sweep: whisper.cpp holds up better under moderate noise, openai/whisper degrades more gracefully under severe noise. Single test clip -- real result, but not a large corpus; treat as directional, not definitive.

2. Raw Engine Performance (isolated, single transcription, cold start)

Metric	whisper.cpp	openai/whisper	Difference
Pure inference time (11s audio)	**285 ms**	326 ms	whisper.cpp 1.14x faster
Real-Time Factor (RTF)	**0.0259 (38.6x faster than real-time)**	0.0296 (33.8x faster than real-time)	whisper.cpp 1.14x faster
Peak memory (RSS)	**372 MB**	763 MB	openai/whisper uses 2.05x more memory
CPU time consumed (user+sys)	**0.23 s**	1.67 s	openai/whisper uses 7.3x more CPU-seconds
Model load time (cold)	**~50-100 ms (within 0.40s total)**	389 ms	whisper.cpp loads faster
Model file size on disk	141.1 MB (ggml-base.en.bin)	139.0 MB (base.en.pt)	Essentially identical

Caveat: this gap is partly a configuration difference, not purely "the engine." whisper.cpp used Metal GPU acceleration on this Mac; the openai/whisper relay runs CPU-only PyTorch (never configured to use MPS/GPU). The CPU-time and memory gap would likely shrink if openai/whisper were explicitly run on the GPU.

3. Live Relay Latency (through the actual app backend, 3 trials each, fully consistent run-to-run)

Metric	whisper.cpp relay	openai/whisper relay
Time to first result	**1.4 s**	3.0 s
Time from last audio sent to final result	2.7 s *after* audio ended	**1.3 s *before* audio fully finished sending**
Messages emitted (11s speech + 2s trailing silence)	11 messages	**3 messages**
Output duplication	Heavy -- same sentence re-emitted 6+ times with growing prefix, by design (sliding-window re-transcription)	**None -- one message per distinct utterance**

Architecturally different strategies: whisper.cpp relay re-decodes a sliding window every 1.2s (fast first feedback, but restates text repeatedly -- client-side dedup needed). openai/whisper relay waits for a detected speech pause (VAD) before transcribing once per utterance (slower first feedback, zero duplication, stops promptly when speech stops).

4. Other factors

Factor	whisper.cpp	openai/whisper
Offline capable	**Yes -- fully local inference**	**Yes -- fully local inference (this is the open-source Python package, not OpenAI's cloud API)**
Internet required	Only once, to download the model file	Only once, to download the model file
GPU acceleration on this Mac	**Yes (Metal)**	No (CPU-only as configured)
Native mobile SDK	**Yes -- official iOS/Android bindings, can run fully on-device with no server**	No -- Python/PyTorch only, needs a server
Streaming API	None natively (relay fakes it via sliding window)	None natively (relay fakes it via VAD segmentation)

Bottom line

If you care most about...	Pick
Raw speed / lowest CPU & memory / mobile embedding	**whisper.cpp**
Clean output with zero duplication / graceful degradation on very noisy audio	**openai/whisper**
Moderate noise accuracy	**whisper.cpp**
Severe noise accuracy	**openai/whisper**